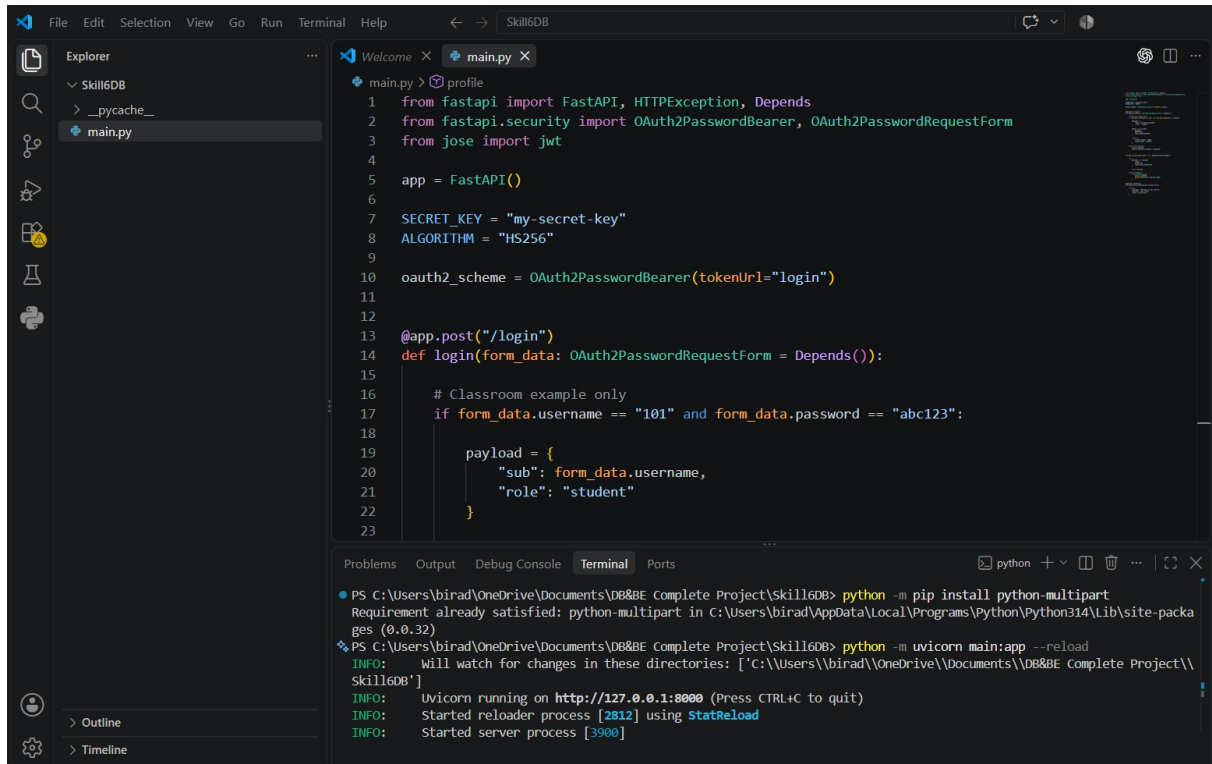


2520090071_Vignesh Kumar

Skill - 6

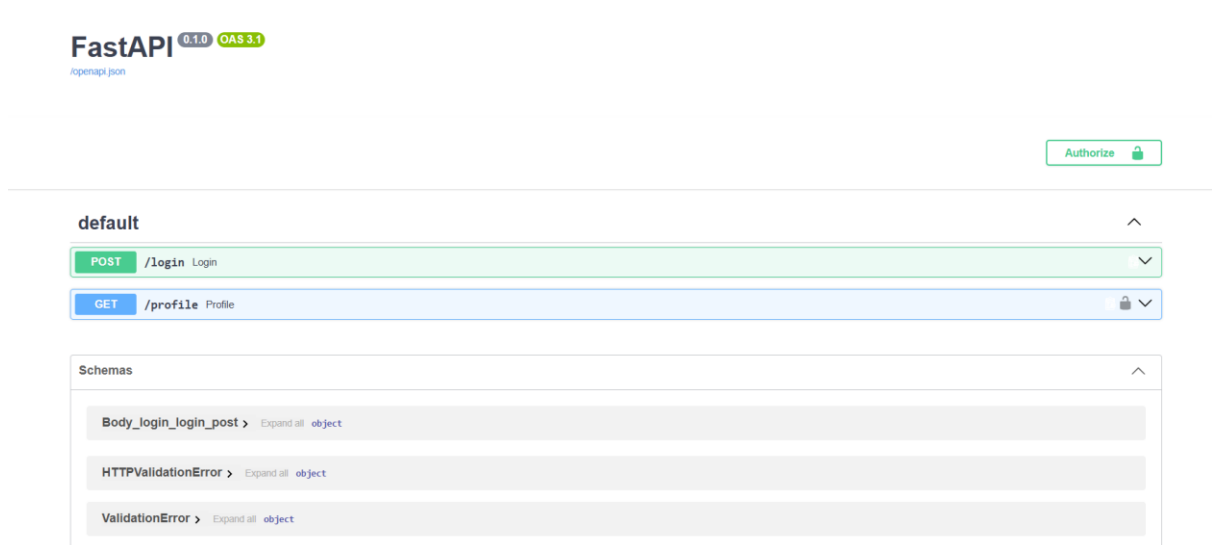


The screenshot shows a Visual Studio Code editor with a project named 'Skill6DB'. The Explorer pane on the left shows the file structure with 'main.py' selected. The main editor displays the code for 'main.py', which is a FastAPI application. The code includes imports for FastAPI, HTTPException, Depends, OAuth2PasswordBearer, OAuth2PasswordRequestForm, and jwt. It defines an app, a secret key, an algorithm, and an OAuth2 scheme. A login endpoint is defined with a POST method, and a profile endpoint is defined with a GET method. The login endpoint uses a form data model and a dependency to validate the user. The profile endpoint returns a JSON response with the user's details.

```
1 from fastapi import FastAPI, HTTPException, Depends
2 from fastapi.security import OAuth2PasswordBearer, OAuth2PasswordRequestForm
3 from jose import jwt
4
5 app = FastAPI()
6
7 SECRET_KEY = "my-secret-key"
8 ALGORITHM = "HS256"
9
10 oauth2_scheme = OAuth2PasswordBearer(tokenUrl="login")
11
12
13 @app.post("/login")
14 def login(form_data: OAuth2PasswordRequestForm = Depends()):
15
16     # Classroom example only
17     if form_data.username == "101" and form_data.password == "abc123":
18
19         payload = {
20             "sub": form_data.username,
21             "role": "student"
22         }
23
```

The terminal at the bottom shows the command to run the application: `python -m uvicorn main:app --reload`. The output indicates that the application is running on `http://127.0.0.1:8000` and that the reloader process is started.

Swagger:



POST /Login:

default

POST /login Login

Parameters

No parameters

Request body required

application/x-www-form-urlencoded

```
grant_type
string | (string |
null)
pattern: ^password$

username * required
string

password * required
string($password)

scope
string

client_id
string | (string |
null)

client_secret
string | (string |
null)($password)
```

Authorize in Swagger:

astAPI 0.1.0 OAS 3.1

Available authorizations

Scopes are used to grant an application different levels of access to data on behalf of the end user. Each API may declare one or more scopes. API requires the following scopes. Select which ones you want to grant to Swagger UI.

OAuth2PasswordBearer (OAuth2, password)

Token URL: login

Flow: password

username:

password:

Client credentials location:

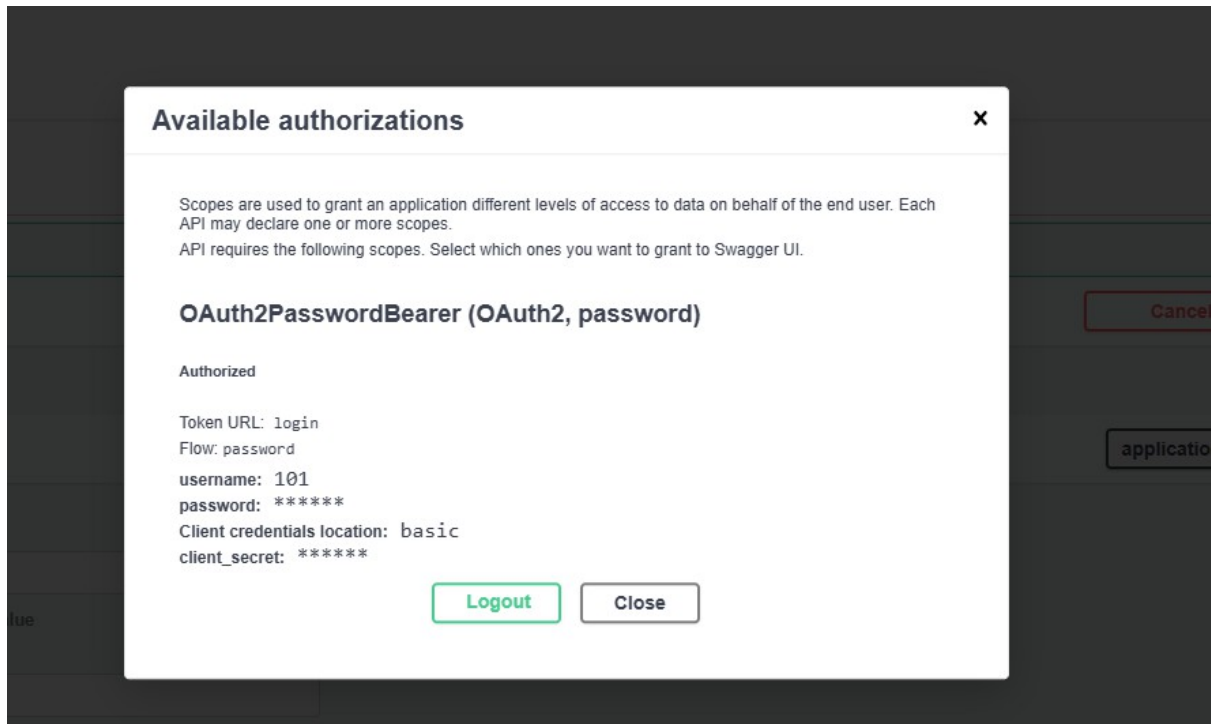
Authorization header

client_id:

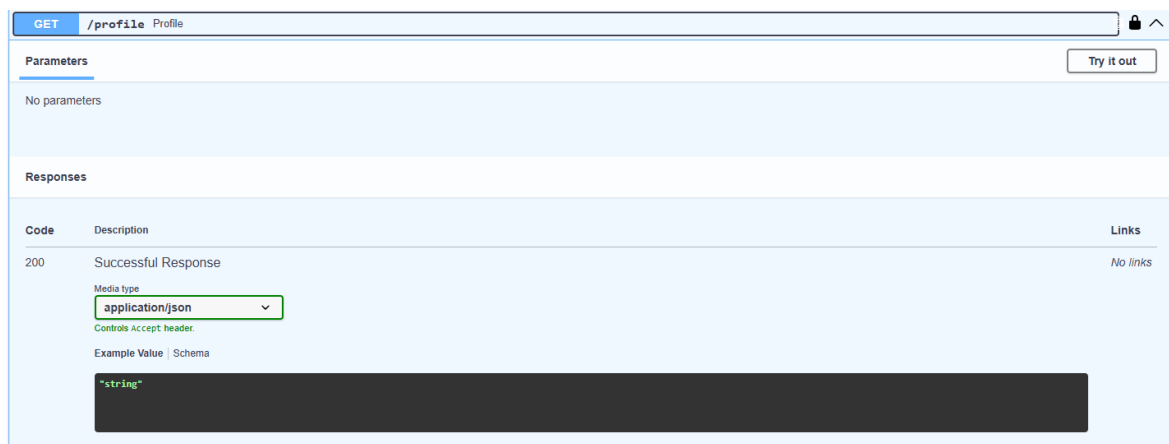
client_secret:

Authorize Close

Send empty value



GET/Profile:



GET

/profile

Profile



Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/profile' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiIxMDE1LjIyY2x1Ijoic3R1ZGVudCJ9.NHqWdLpBiQ5BFQ1iKadoWVLOjblUPM8m4aRQGMK1zI'
```

Request URL

http://127.0.0.1:8000/profile

Server response

Code	Details
200	Response body <pre>{ "message": "Welcome to your profile", "user_id": "101", "role": "student" }</pre> Download Response headers <pre>content-length: 70 content-type: application/json date: Fri, 18 Sep 2026 04:07:00 GMT server: uvicorn</pre>

Responses

Code	Description	Links
------	-------------	-------